# Recovering a time-dependent control field from qubit expectation values: physics-informed neural networks versus classical inversion

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**Abstract**

We study the inverse problem of recovering an unknown time-dependent control field $f(t)$ of a driven qubit, $H = \frac{1}{2}\sigma_z + \frac{1}{2}f(t)\sigma_x$, from measured expectation values, and compare physics-informed neural networks (PINNs) with classical inversion under a single protocol: identical data, initial state $|0\rangle$, no access to the true field, five seeds, and two metrics (field error and re-simulated trajectory error). PINNs trained with Adam alone, as in common practice, converge to an over-smoothed field with $\sim 25\%$ error regardless of how the field is parametrized (neural network, Fourier series, B-splines, MixFunn). A Levenberg–Marquardt stage reduces the error by two orders of magnitude (to $\sim 0.2\%$), which shows that the optimizer, not the representation, is the bottleneck. With complete, noiseless measurements, classical linear least squares and shooting with a suitable basis remain about ten times more accurate and thousands of times faster [**TODO:** update with the complete benchmark and the noisy case]. We also show why the initial state $|+\rangle$, an eigenstate of the control operator, makes the field unidentifiable at early times.

## 1 Introduction

Physics-informed neural networks (PINNs) [1] have been used for quantum control [2] and for learning time-independent Hamiltonian parameters [3]. Here we study the inverse problem of recovering an unknown *time-dependent* control field from the measured expectation values of a driven qubit, and we ask a simple question: under a single, fair protocol, when do PINNs beat classical inversion? Recovering a control field from measurements is relevant for calibrating and characterizing quantum devices, where the pulse that actually reaches the qubit differs from the programmed one [**TODO:** add references on pulse characterization]. Earlier PINN studies of this kind of problem compare a single model against the ground truth. Here we compare several field parametrizations and optimizers against classical estimators that use the same data. Section 2 states the problem and analyses its identifiability, Sec. 3 describes the methods and the protocol, Sec. 4 presents the results, and Sec. 5 concludes.

Contributions:

- a single protocol (same data, same initial state, no access to the true field, 5 seeds, two metrics) that compares four field parametrizations—neural network, periodic Fourier series, cosine series and B-splines, plus MixFunn [4]—and classical inversion (direct, linear least squares, shooting);

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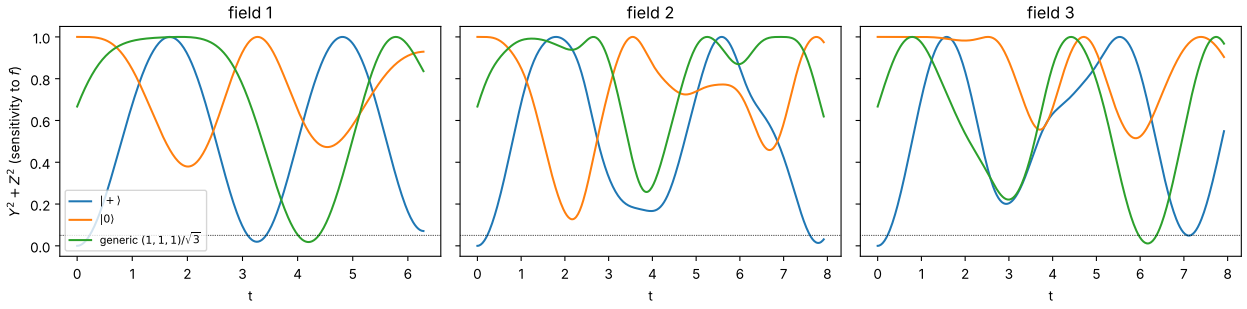



Figure 1: $Y^2 + Z^2$, the sensitivity of the data to the field, for three initial states. Produced by `run/diagnostico.py`.

- an identifiability analysis that explains why the choice of initial state matters (Sec. 2.1);
- evidence that the optimizer, not the parametrization, limits PINN accuracy here: a Levenberg–Marquardt (Gauss–Newton) stage improves the field error by one to two orders of magnitude over Adam, in line with recent work on second-order PINN training [5–9];
- a spectral-resolution study: the error as a function of the frequency of the field [**TODO:** run].

## 2 Problem

The qubit evolves under

$$H(t) = \tfrac{1}{2}\,\sigma_z + \tfrac{1}{2}\,f(t)\,\sigma_x, \tag{1}$$

and the Bloch vector $\boldsymbol{r} = (X, Y, Z) = (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$ obeys

$$\dot{X} = -Y, \qquad \dot{Y} = X - fZ, \qquad \dot{Z} = fY. \tag{2}$$

Given $N = 100$ equally spaced measurements of $\boldsymbol{r}(t_i)$ on $[0, T]$, possibly with Gaussian noise of standard deviation $\sigma$, we want $f(t)$. Test fields: (1) $f = \cos t$, $T = 2\pi$; (2) $f = 1.5\sin 2t + 0.8\cos(3.5t + 0.5) + 0.4\sin 7t$, $T = 2.52\pi$; (3) $f = [1.5\sin 2t - 0.3\cos(3t - 0.5) + 0.4\sin 7t]\,\sin^2(\pi t/T)$, $T = 2.52\pi$; and, for the spectral study, single-tone pulses $f = \sin(\omega t)\sin^2(\pi t/T)$.

### 2.1 Identifiability and choice of the initial state

The field enters Eq. (2) only through $fZ = X - \dot{Y}$ and $fY = \dot{Z}$, so the information on $f(t)$ carried by the data is proportional to $Y^2 + Z^2 = 1 - X^2$. The state $|+\rangle$, an eigenstate of the control operator $\sigma_x$, gives $Y^2 + Z^2 = 0$ at $t = 0$: there the field has no effect on the state and cannot be identified. With $|0\rangle$, $\dot{Y}(0) = -f(0)$ and $Y^2 + Z^2 \geq 0.13$ over the whole interval for the three test fields (Fig. 1). We therefore use $|0\rangle$ throughout. With the same measurement budget this halves the error of the classical estimators and reduces its spread across noise realizations (from $0.15 \pm 0.05$ to $0.07 \pm 0.01$ for linear least squares at $\sigma = 0.01$, five noise realizations; `data/parte1/resultados_desenho_experimental.csv`). With only $\langle\sigma_z\rangle$ measured, however, $|0\rangle$ makes $f$ and $-f$ indistinguishable, because $\sigma_z H(f)\sigma_z = H(-f)$.

## 3 Methods

**Field parametrizations.** (i) Fourth output of the state network; (ii) periodic Fourier series with period $T$ and $M = 10$ harmonics; (iii) cosine series $\sum_{n=0}^{40} c_n \cos(n\pi t/T)$, which has no Gibbs phenomenon when $f(0) \neq f(T)$; (iv) cubic B-splines with 80 uniform basis functions; (v) MixFunn [4].

**PINN.** A network $\mathrm{NN}(s)$ with input $s = 2t/T - 1$ represents the state, with the initial condition imposed exactly, $\boldsymbol{r}(t) = \boldsymbol{r}_0 + (t/T)\,\mathrm{NN}(s)$. When the pulse is known to start and end at zero (field 3 and the spectral pulses), the field is constrained exactly as well, $f = g - (1-\tau)g(0) - \tau g(T)$ with $\tau = t/T$ and $g$ the unconstrained parametrization [10]. The loss is the sum of the mean squared residual of Eq. (2) over collocation points, the mean squared data misfit and, for noisy data, a smoothness penalty $\lambda_f T^{-1} \int f''^2\, dt$. Collocation points are the data points plus 200 uniform random points redrawn at every Adam epoch. Time derivatives are computed in forward mode inside the network. The coefficients of linear parametrizations are initialized by linear least squares on the data.

**Optimization.** 3000 Adam steps (cosine schedule), followed by L-BFGS, dense BFGS, or Levenberg–Marquardt (LM) with Nielsen's damping update [11]. Because the loss is exactly $\|\boldsymbol{\rho}(\theta)\|^2$ for a residual vector $\boldsymbol{\rho}$, the LM step is a Gauss–Newton step in the sense of [8, 9]; when parameters outnumber residuals it is solved in residual space [12]. We also tested causal weights [13], gradient-norm loss balancing [14] and variable projection.

**Classical baselines.** Direct pointwise inversion $f = [Z(X - \dot{Y}) + Y\dot{Z}]/(Y^2 + Z^2)$ with spline derivatives; linear least squares for the coefficients of a basis; and shooting: Bloch-equation integration with sensitivities and LM on the data misfit, initialized by least squares.

**Protocol.** Identical data for all methods per (field, noise, seed); initial state $|0\rangle$; the true field never enters a loss or a hyper-parameter choice (regularization weights are fixed a priori by the noise level); five seeds; two metrics: relative $L_2$ error of the field and the RMS error of the trajectory re-simulated with the recovered field. Code: `src/`, `run/rodar.py`.

## 4 Results

[**TODO:** Preliminary: 260 of 440 runs. Missing: PINN+LM with noise; Adam-only and L-BFGS recipes for the network, Fourier, cosine and B-spline fields. Update every number below from `data/resultados/padrao/RESULTADOS.md` when complete.]

**Adam-only training.** PINNs trained as in common practice (small network, raw time input, residual only at the data points, Adam with a fixed learning rate) recover field 3 with $\approx 25\%$ error for every field parametrization. The recovered field is a smoothed version of the true one, while the re-simulated trajectory is already close to the data (RMS error $\sim 10^{-2}$): a small trajectory error does not imply an accurate field.

**Second-order optimization.** Adding a Levenberg–Marquardt stage after 3000 Adam steps reduces the median field error for field 3 to 0.20% (field as network output), 0.18% (cubic B-splines with 80 functions) and 0.19% (cosine series), consistently over five seeds (Fig. 2). For field 1 ($\cos t$) all linear parametrizations reach $\sim 10^{-5}$. Once the optimizer is adequate the parametrization

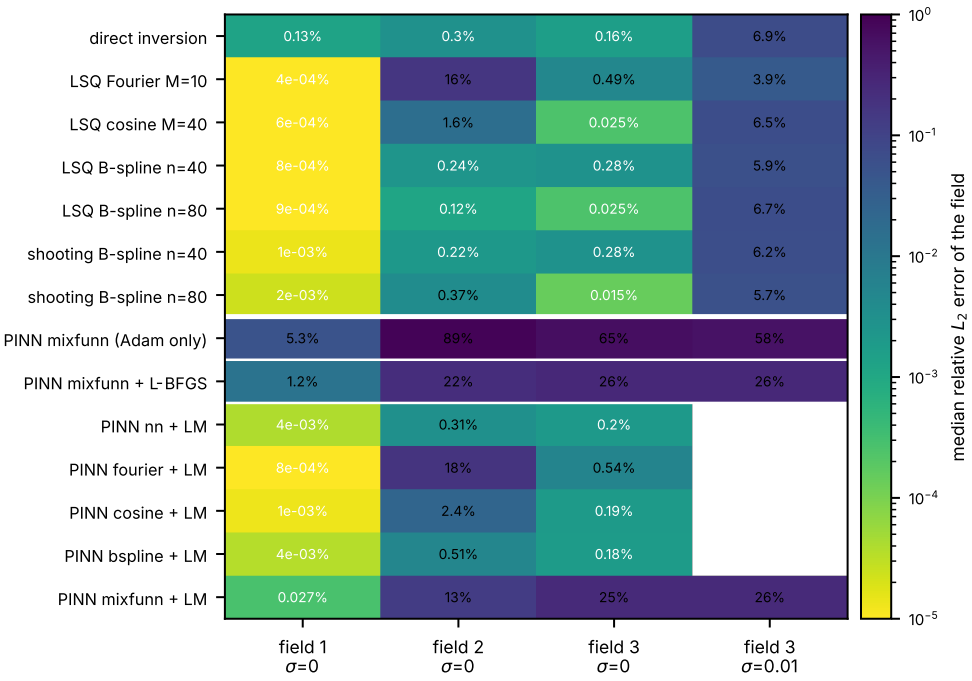



Figure 2: Median relative error of the recovered field over five seeds, for every method and scenario (`run/fig_artigo.py`, data in `data/resultados/padrao`).

matters: the periodic Fourier basis cannot represent field 2, for which $f(0) \neq f(T)$ (18%, Gibbs phenomenon), while the network reaches 0.31%. MixFunn reaches $3 \times 10^{-4}$ for $\cos t$, but it did not converge within the time limit for the other fields (13–25%).

**Comparison with classical inversion.** With complete, noiseless measurements, classical estimators with the same basis are more accurate than the best PINN: for field 3, shooting with 80 B-splines reaches 0.015% and linear least squares 0.025%, against $\approx 0.2\%$ for the PINNs, at a cost below one second instead of minutes. Enlarging the basis (40 to 80 B-splines; cosines instead of a periodic Fourier series) improves the classical estimators by a factor of ten. With noise ($\sigma = 0.01$) all classical estimators lie between 3.9% and 6.9%, and the information content of the data, not the method, limits the accuracy [**TODO:** PINN results with noise].

[**TODO:** Ablation (figure commented out): run profile `ablacao`. Spectral resolution: profile `espectral` not run yet.]

## 5 Conclusions

(Preliminary.) Under a single protocol, the accuracy of PINNs for this inverse problem is limited by the optimizer: Adam alone yields an over-smoothed field, and a Gauss–Newton (Levenberg–Marquardt) stage improves it by two orders of magnitude. With complete, noiseless data, classical least squares and shooting with a good basis are still more accurate and much cheaper, so the case for PINNs has to be made where these estimators do not apply directly: partial observations (e.g. only $\langle \sigma_z \rangle$, where $|0\rangle$ makes the sign of $f$ unidentifiable), several initial states sharing one field, and open-system dynamics [**TODO:** ].

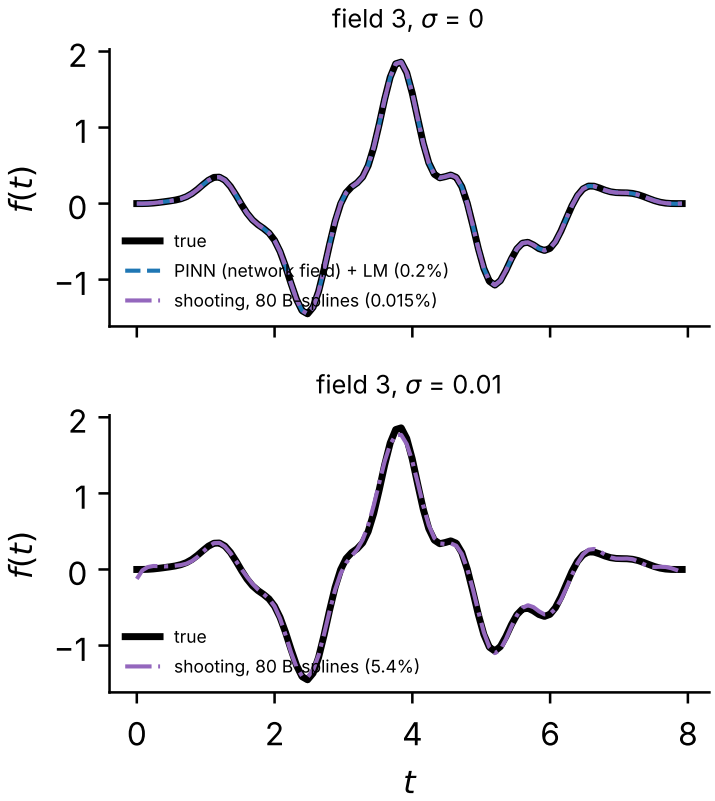



Figure 3: Field 3 recovered by an Adam-only PINN, a PINN with an LM stage and shooting (seed 0).

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